Is it all downhill after 20? Exploring age and gender differences in language skills

Introduction: Psycholinguistic research has long relied on a narrow participant pool: predominantly female undergraduate students. As a result, comparatively little is known about language skills in middle-aged adults and in men across the lifespan. This is particularly relevant as a female advantage has been shown in at least some language skills (Kljajevic et al., 2023; Scheuringer et al., 2017). This raises questions about the extent to which findings and theories based on traditional psycholinguistic samples generalize to the broader adult population**.** We explore this concern using a battery of linguistic and other cognitive tests, asking (a) if male and female participants differ in performance on linguistic and domain-general skills, (b) similarly, if there are age-related performance differences in these domains.

Methods: 863 participants aged 18 to 40 completed 33 tests of linguistic and domain-general skills. Participants were native speakers of Dutch and included 609 women and 252 men (other gender identities excluded due to small sample size). We analyzed five tests of visual and auditory processing speed, four working memory tests (forward and backward digit spans and Corsi spans), five measures of linguistic knowledge (author recognition, picture vocabulary, prescriptive grammar, spelling and antonym knowledge), four word production tests (picture naming, rapid automatized naming, categorical and phonemic verbal fluency) and three word comprehension tests (lexical decision, rhyme judgement and semantic categorization).

Results: A confirmatory factor analysis (CFA) was fit to the five domains, and the model was assessed for measurement invariance. Metric invariance was established, indicating that the relationship between the tests and the latent constructs was the same for the two genders. However, a difference in intercepts for the tests in the linguistic knowledge domain revealed measurement-level differences. This confirms what we see in the data: author recognition scores are higher for women, but picture vocabulary scores are higher for men (see Figure 1).

Structural equation modeling (SEM) is currently underway and preliminary results suggest that while men have significantly higher latent means in working memory, women have greater latent ability for word production and comprehension. Higher age significantly predicts lower working memory and processing speed scores but has no effect on the language domains and these effects are not influenced by gender (Figure 2). This can also be seen in the histograms: women show higher scores on RAN, semantic verbal fluency, and the word comprehension tests (see Figure 3). Unlike the measurement invariance results, this indicates differences in latent ability.

Discussion: With respect to age, modest declines in processing speed and working memory in middle adulthood could, in principle, exert downstream effects on language performance. However, the present findings offer an overall optimistic picture: we observe no evidence for meaningful decline in language skills or for a gender difference in age effects.

The interpretation of gender effects on language ability appears more nuanced. Measurement-level disparities detected by the measurement invariance models suggest that tasks such as author recognition and vocabulary are sensitive to gender-related influences, potentially indicating that some items (author names or vocabulary words) are more readily recognized by one gender than another. The SEM results, however, suggest meaningful latent mean differences between men and women in word production and comprehension skills, with implications for the generalizability of prior findings to the broader population. These results underscore the importance of recruiting more diverse participant samples in experimental psycholinguistics.

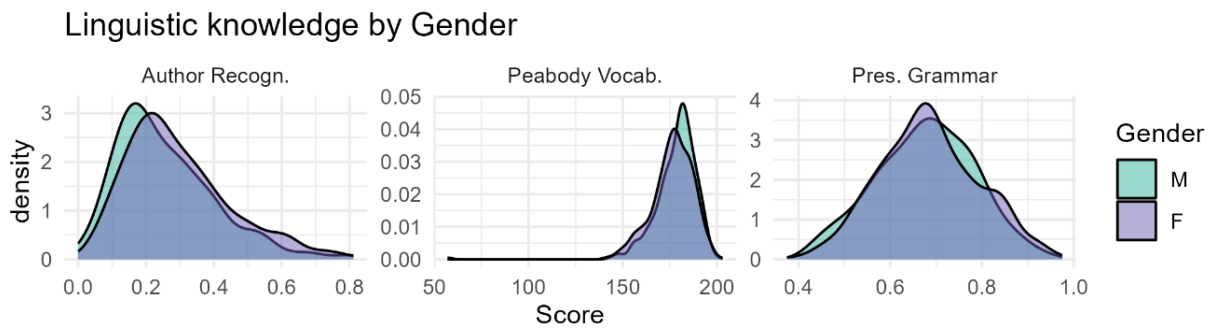

Figure 1: Effect of gender on selected linguistic knowledge tests, across all ages.

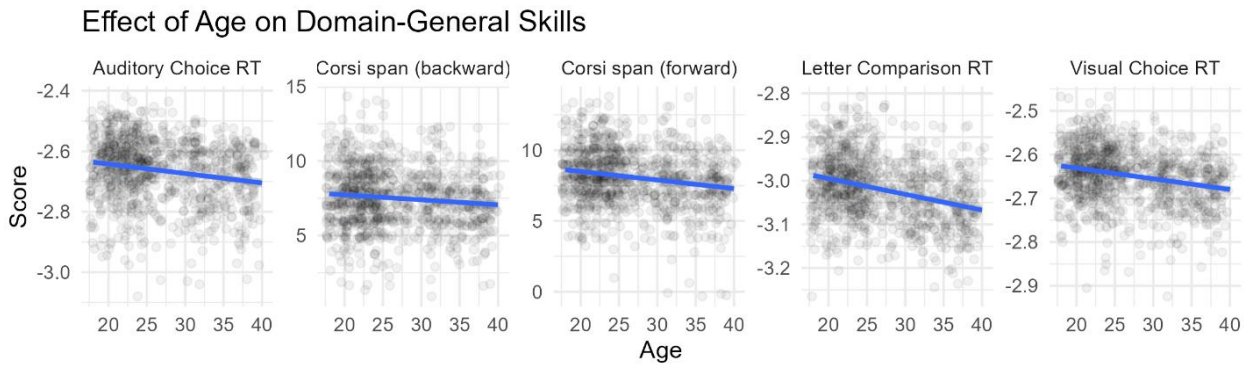

Figure 2: Effect of age on domain-general skills; no effects found on language skills. Note that RT tasks are shown on RTs that have been log-transformed and multiplied by -1 so that a higher score indicates “better” performance.

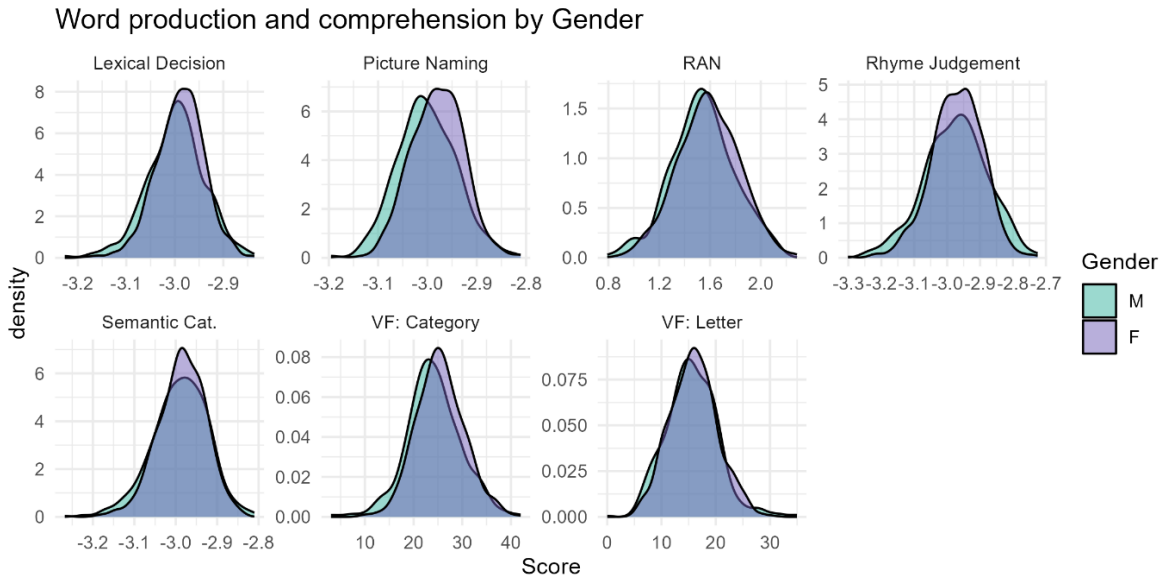

Figure 3: Effect of gender on word production and comprehension tests, across all ages.

References:

Kljajevic, V., Evensmoen, H. R., Sokołowski, D., Pani, J., Hansen, T. I., & Håberg, A. K. (2023). Female advantage in verbal learning revisited: A HUNT study. Memory, 31(6), 831-849.

Scheuringer, A., Wittig, R., & Pletzer, B. (2017). Sex differences in verbal fluency: The role of strategies and instructions. Cognitive processing, 18(4), 407-417.